

Reproducing FreeZe: Training-Free Zero-Shot 6D Pose Estimation with Foundation Models

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Abstract

I reproduced FreeZe [1], a training-free approach to 6D object pose estimation leveraging frozen foundation models (GeDi, DINOv2). During reproduction, I identified and resolved critical implementation challenges, including camera intrinsics consistency, mesh normalization mismatches, and feature matching ambiguity in cluttered scenes. I introduce a microservice-based architecture with SAM2-based zero-shot segmentation, achieving end-to-end 6D pose estimation. My analysis reveals that while the core approach is sound, feature matching in highly ambiguous scenarios (e.g., 18+ similar objects) remains fundamentally challenging without additional geometric priors or learned reranking.

1 Introduction

FreeZe addresses 6D object pose estimation (predicting rotation and translation) without task-specific training. The key insight is combining complementary foundation models: GeDi (3D geometric descriptor) and DINOv2 [2] (2D visual encoder). By fusing these modalities and applying RANSAC-based registration, the method generalizes to unseen object categories.

While the original FreeZe paper provides the theoretical foundation, my work focuses on experimenting with a different architectural approach to implementation. This work documents the findings from building an improved, deployable architecture from the ground up.

2 FreeZe Pipeline Overview

The FreeZe pipeline comprises four main stages.

1. Query Processing: Points are sampled from the 3D mesh and encoded with 32-dim GeDi features. The object is then rendered from multiple viewpoints, and 1024-dim DINOv2 features are extracted from each render, back-projected to the 3D points, and averaged. **2. Target Processing:** The RGB-D image is segmented (I use SAM2 [3] instead of the original CNOS), the depth is lifted to a 3D point cloud, and features are extracted and fused as in the query branch. **3. Feature Fusion:** The L2-normalized geometric and visual features are concatenated: $\mathbf{f} = \left[\frac{\mathbf{v}}{\|\mathbf{v}\|_2} \parallel \frac{\mathbf{g}}{\|\mathbf{g}\|_2} \right]$. **4. Pose Estimation:** Query points are matched to target points via nearest neighbors in the fused feature space, and RANSAC is applied to estimate the final transformation (R, t) .

3 Key Challenges & Solutions

3.1 Challenge 1: Camera Intrinsics Mismatch

A primary challenge was inconsistent camera parameter handling. My initial auto-estimated intrinsics ($f_y \approx 935$) conflicted with the HOPE [4] dataset’s ground-truth intrinsics ($f_x = 1390.5, f_y = 1387$). This mismatch between the pose estimation and rendering coordinate systems caused severe visual artifacts, with

mesh overlays misaligned by over 100 pixels. The solution was to load intrinsics from scene metadata and propagate them consistently through all pipeline stages, ensuring geometric and visual consistency.

3.2 Challenge 2: Mesh Normalization Inconsistency

Similarly, feature statistics were misaligned due to incorrect mesh normalization. GeDi expects point clouds normalized such that their bounding box diagonal is 0.2m. My initial renderer used a different scaling method. I resolved this by implementing the correct diagonal-based scaling, to match GeDi’s training distribution.

3.3 Challenge 3: Feature Matching Ambiguity

Switching to a zero-shot segmenter (SAM2) for generality created proposal scarcity in cluttered, 18-object scenes. With few proposals, one-way nearest-neighbor matching produced many spurious correspondences, leading RANSAC to converge on locally consistent but globally incorrect poses. To mitigate this, I implemented a series of solutions: (1) increasing SAM2 proposals by adjusting `points_per_side` from 32 to 64; (2) enforcing tighter matching criteria with a `ratio_test` of 0.80 and mutual nearest-neighbor filtering; (3) applying geometric constraints like depth filtering; and (4) using a stricter RANSAC inlier threshold of 5mm. Collectively, these changes reduced false matches by $\sim 40\%$ and improved correct proposal selection from 30% to 71%. However, this highlights that the matching task remains fundamentally ambiguous in scenes with many similar objects.

4 Architectural Contributions

4.1 Microservice-Based Design

Unlike monolithic implementations, I adopted a design using isolated subprocesses for each foundation model (GeDi, DINOv2, SAM2). This modular architecture prevents dependency conflicts (e.g., differing CUDA requirements), allows for easy swapping of models, and prepares the system for future distributed deployment.

4.2 SAM2 vs. CNOS

The original CNOS requires pre-rendered templates for segmentation. My use of SAM2’s automatic mask generator provides class-agnostic proposals, making the pipeline **truly zero-shot**. This trades the specificity of template matching for generality, though it can result in fewer and less discriminative initial proposals, as discussed in the challenges.

5 Results & Validation

My pipeline successfully processes HOPE dataset scenes end-to-end, generating accurate 6D poses visualized with 3D mesh overlays. Correctness was validated via ground-truth pose injection, which produced perfect alignment.

6 Conclusion

I successfully reproduced FreeZe and documented critical implementation details for deploying a robust system. The core concept of leveraging foundation models for zero-shot pose estimation is sound, but its practical application is challenged by feature ambiguity in cluttered environments. Future work should explore learned proposal reranking, and evaluation on additional datasets such as BOP and YCB. My implementation is available at [\[github.com/crisiumnih/freeze-pipeline\]](https://github.com/crisiumnih/freeze-pipeline).

References

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